

Coarse-to-Fine Image Classification with Machine Learning Approaches and Deep Visual Features

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1 Introduction

This report investigates two image classification tasks under a coarse-to-fine setting. Task 1 focuses on coarse-grained classification of 10 visually distinct animal categories, while Task 2 addresses fine-grained classification of 10 visually similar bird species. The progression from Task 1 to Task 2 enables investigation of how feature representations and classification models behave as inter-class visual similarity increases.

In addition to the provided image descriptors and engineered features, deep visual embeddings extracted from a pretrained ResNet18 model were evaluated, both independently and in combination with handcrafted features. Multiple supervised learning algorithms were compared using stratified cross-validation, followed by hyperparameter tuning and holdout evaluation of the strongest models.

Overall, deep visual embeddings substantially improved classification performance, particularly for the fine-grained task, while combining handcrafted and deep features produced the strongest feature representation. The experiments further demonstrate the increased difficulty of fine-grained classification and provide insight into how feature representation, classifier choice and class similarity influence predictive performance.

2 Methodology

2.1 Overview

Two image classification tasks were investigated under a coarse-to-fine classification setting. Both tasks followed a similar machine learning pipeline consisting of feature extraction, data preprocessing, feature engineering, model training, hyperparameter tuning and performance evaluation. Multiple supervised learning algorithms were explored to investigate how different models perform under increasing classification difficulty.

2.2 Dataset Description

Table 1: Summary of the datasets used for the two classification tasks.

Dataset Characteristic	Task 1: Coarse-grained	Task 2: Fine-grained
Classes	Bird, Butterfly, Cat, Deer, Dog, Elephant, Frog, Horse, Sheep, Spider	American Goldfinch, Blue Jay, Cardinal, Herring Gull, House Sparrow, Red-winged Blackbird, Ring-billed Gull, Song Sparrow, Wilson Warbler, Yellow Warbler
Number of classes	10	10
Training images	3,750	417
Test images	1,250	180
Combined features	771	787

Table 1 summarises the datasets used for the two classification tasks. Task 2 represents the more challenging fine-grained setting, with substantially fewer training images and greater visual similarity between classes.

2.3 Feature Representation

Three feature configurations were evaluated for both classification tasks:

1. Provided and engineered image features;
2. Deep visual embeddings extracted using a pretrained ResNet18 model; and
3. A combined representation containing both the provided and engineered features and ResNet18 embeddings.

The provided dataset already contained handcrafted image descriptors derived from the images. Additional preprocessing and feature engineering steps were applied before model training. Missing values were replaced with zeros to ensure consistent feature dimensions across all instances. Features from different sources were merged using image identifiers to create unified training and test representations.

A pretrained ResNet18 model[1] was used as a fixed feature extractor, with its final classification layer removed to obtain deep visual embeddings. Images were resized and normalised using the standard ImageNet preprocessing pipeline. These embeddings provide higher-level representations of visual characteristics such as shape, texture and structural patterns that may not be adequately captured by handcrafted descriptors.

Feature comparison experiments were conducted to evaluate the contribution of handcrafted features, ResNet embeddings and their combination across both tasks. Due to computational cost, LightGBM was used as the primary comparison model since it provided a strong balance between efficiency and classification performance.

2.4 Machine Learning Models

Several supervised learning algorithms were evaluated to represent different types of decision boundaries and ensemble strategies. Logistic Regression was included as a linear baseline to assess how well the

extracted features separated classes without complex non-linear modelling. Random Forest and Extra Trees were used to capture non-linear relationships and interactions between the high-dimensional image features through tree-based ensemble learning.

LightGBM[2], XGBoost[3] and CatBoost[4] were explored as boosting-based approaches capable of modelling complex feature interactions. An RBF SVM was also evaluated because its non-linear kernel can model complex class boundaries in the combined feature space. Since RBF kernels depend on distances between observations, features were standardised before training, while PCA was used to reduce dimensionality and redundant variation.

Soft-voting ensembles were additionally investigated by combining probability predictions from multiple high-performing classifiers. This was intended to test whether classifiers with different learning mechanisms could provide complementary predictions and improve robustness.

2.5 Hyperparameter Tuning

After the initial model comparison, hyperparameter tuning was conducted on the strongest-performing model for each task. For Task 1, tuning focused on the RBF SVM regularisation parameter C , which controls the trade-off between maximising the classification margin and penalising training errors. For Task 2, the Extra Trees classifier was tuned by varying the number of trees (`n_estimators`) and the minimum leaf sample size (`min_samples_leaf`), which influence ensemble stability and tree complexity, respectively. Hyperparameter configurations were evaluated using stratified cross-validation, and the configuration with the highest mean validation accuracy was selected for final evaluation.

2.6 Evaluation Strategy

Model performance was evaluated using stratified 5-fold cross-validation to maintain class distributions across folds. Accuracy was used as the primary model-selection metric because the datasets were approximately balanced across the 10 classes and all classification errors were treated equally. Mean cross-validation accuracy and standard deviation were therefore used to compare predictive performance and stability across folds.

Precision, recall and F1-score were additionally examined at the class level to identify differences in performance that overall accuracy could obscure. A stratified holdout validation set was used for final evaluation, with confusion matrices and confused class pairs analysed to identify systematic errors, particularly among visually similar categories.

Finally, the selected models were retrained on the full training data using the chosen hyperparameters and used to generate predictions for the Kaggle test sets.

3 Results

3.1 Evaluation Metrics

Model comparison and hyperparameter tuning were conducted using stratified 5-fold cross-validation, with mean accuracy and standard deviation reported across folds. The selected models were subsequently evaluated on a stratified holdout validation set using accuracy, precision, recall and F1-score. Confusion matrices and frequently confused class pairs were further examined to identify systematic classification errors.

3.2 Task 1: Coarse-Grained Animal Classification

3.2.1 Feature Representation Comparison

Table 2: Task 1 cross-validation performance across feature representations.

Feature Representation	Mean CV Accuracy
Provided + engineered	60.59%
ResNet18	83.52%
Combined	85.36%

The provided and engineered features achieved a mean cross-validation accuracy of 60.59%, compared with 83.52% using ResNet18 embeddings. Combining both representations produced the highest mean accuracy of 85.36%. The combined representation was therefore used for subsequent Task 1 model comparisons.

3.2.2 Model Comparison

Multiple supervised learning models were implemented and evaluated using the combined feature representation.

Table 3: Task 1 cross-validation performance across classification models.

Model	Mean CV Accuracy	Std CV Accuracy
RBF SVM	87.28%	1.57%
Soft-voting Ensemble	87.15%	0.53%
XGBoost	85.97%	0.31%
LightGBM	85.60%	0.15%
Logistic Regression	85.25%	2.19%
CatBoost	84.08%	1.07%
Random Forest	83.60%	1.20%

The RBF SVM achieved the highest mean cross-validation accuracy of 87.28%, followed by the soft-voting ensemble at 87.15%. XGBoost, LightGBM, Logistic Regression, CatBoost and Random Forest achieved mean cross-validation accuracies of 85.97%, 85.60%, 85.25%, 84.08% and 83.60%, respectively. The RBF SVM was selected for subsequent hyperparameter tuning.

3.2.3 Hyperparameter Tuning

After identifying the RBF SVM as the strongest-performing model, hyperparameter tuning was conducted on the regularisation parameter C.

Table 4: Task 1 RBF SVM hyperparameter tuning results.

C	Mean CV Accuracy	Std CV Accuracy
1	87.09%	1.37%
5	87.25%	1.62%
8	87.28%	1.57%
15	87.28%	1.57%

The highest mean cross-validation accuracy of 87.28% was achieved at both $C = 8$ and $C = 15$. Based on the tuning procedure, $C = 15$ was selected for the final RBF SVM model.

3.2.4 Final Validation Performance

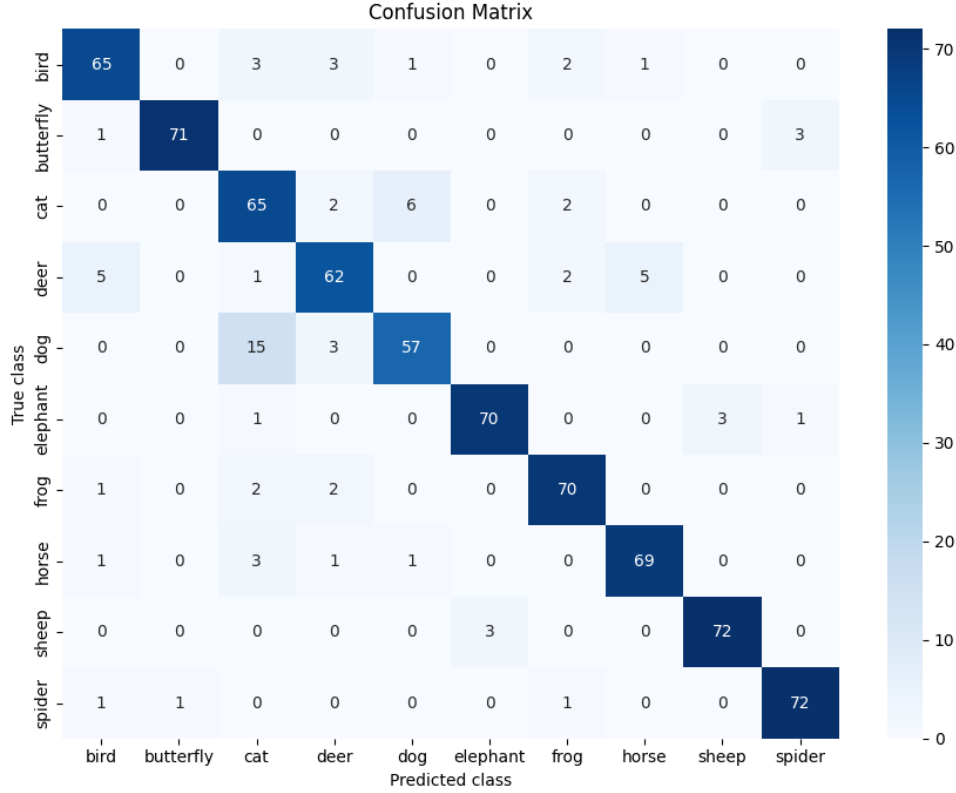


Figure 1: Confusion matrix for the final RBF SVM on the Task 1 holdout validation set.

The final RBF SVM achieved a holdout validation accuracy of 89.73%. The highest number of misclassifications occurred between dog and cat, with 15 dog images classified as cat and 6 cat images classified as dog. For deer, 5 images were classified as horse and 5 as bird. Figure 1 presents the corresponding confusion matrix.

3.3 Task 2: Fine-Grained Bird Classification

3.3.1 Feature Representation Comparison

Table 5: Task 2 cross-validation performance across feature representations.

Feature Representation	Mean CV Accuracy
Provided + engineered	32.13%
ResNet18	85.38%
Combined	86.34%

The provided and engineered features achieved a mean cross-validation accuracy of 32.13%, while the ResNet18 embeddings achieved 85.38%. The combined representation achieved the highest mean accuracy of 86.34% and was used for the subsequent Task 2 model comparison.

3.3.2 Model Comparison

Table 6: Task 2 cross-validation performance across classification models.

Model	Mean CV Accuracy
Extra Trees	86.34%
Soft-voting Ensemble	85.61%
CatBoost	84.91%
LightGBM	83.69%

Extra Trees achieved the highest mean cross-validation accuracy of 86.34%, followed by the soft-voting ensemble at 85.61%, CatBoost at 84.91% and LightGBM at 83.69%. Extra Trees was selected for subsequent hyperparameter tuning.

3.3.3 Hyperparameter Tuning

Table 7: Task 2 Extra Trees hyperparameter tuning results.

n_estimators	min_samples_leaf	Mean CV Accuracy	Std CV Accuracy
500	2	86.34%	3.65%
800	2	86.10%	2.30%
800	1	85.62%	3.33%
500	1	85.62%	2.69%
300	2	85.62%	3.60%
300	3	84.42%	1.24%
800	3	84.42%	2.11%
500	3	84.18%	1.73%
300	1	83.70%	2.55%

Extra Trees was evaluated across different combinations of `n_estimators` and `min_samples_leaf`. The highest mean cross-validation accuracy of 86.34% was obtained with `n_estimators`=500 and `min_samples_leaf`=2. This configuration was selected for the final Extra Trees model.

3.3.4 Final Validation Performance

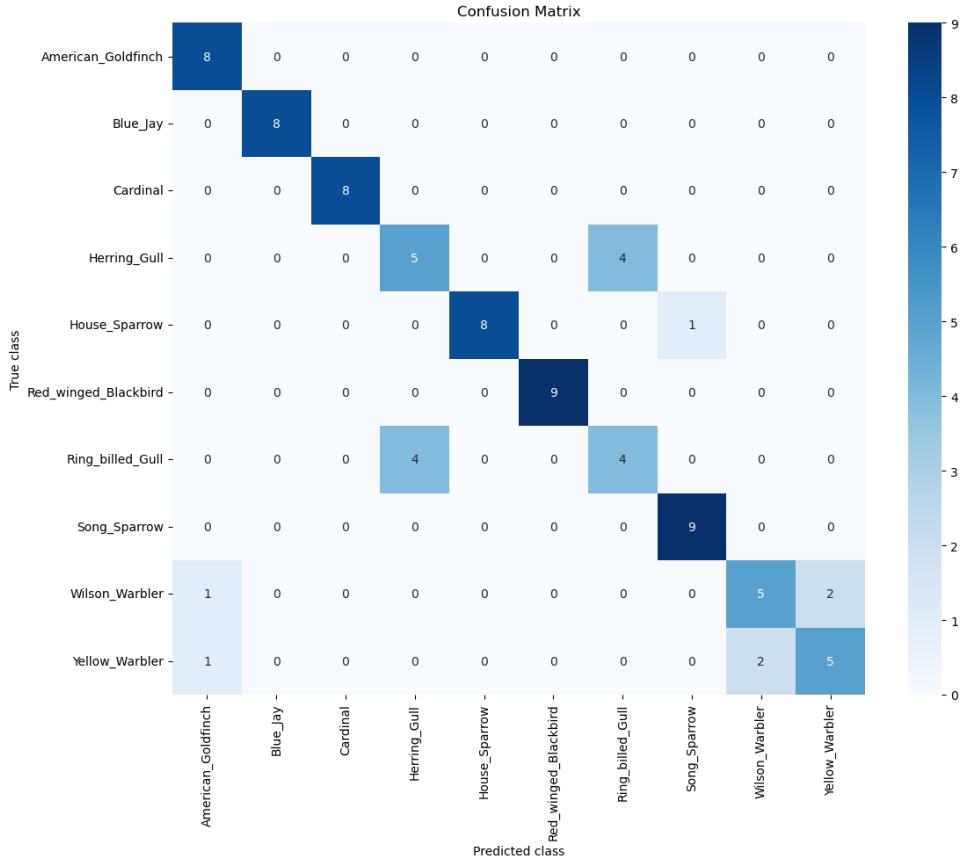


Figure 2: Confusion matrix for the final Extra Trees classifier on the Task 2 holdout validation set.

The tuned Extra Trees classifier achieved a holdout validation accuracy of 82.14%. Blue Jay, Cardinal and Red-winged Blackbird were classified correctly across all validation samples. The largest confusion occurred between Herring Gull and Ring-billed Gull, with 4 misclassifications in each direction. Wilson Warbler and Yellow Warbler were also confused, with 2 misclassifications in each direction. Figure 2 presents the corresponding confusion matrix.

4 Discussion

4.1 Effect of Feature Representation

The experiments demonstrated that feature representation had a major impact on classification performance across both tasks. In particular, the ResNet18 embeddings consistently outperformed the provided and engineered features, with the improvement being especially significant for the fine-grained bird classification task. While handcrafted features achieved moderate performance for Task 1, they performed poorly on Task 2, indicating that they were insufficient for capturing subtle visual differences between highly similar bird species.

The strong performance of the ResNet18 embeddings is consistent with their ability to capture higher-level semantic image information such as texture, shape, colour distribution and structural patterns. Since the model was pretrained on large-scale image datasets, the extracted embeddings

provided more robust and transferable visual representations than handcrafted features alone. Combining the handcrafted and ResNet18 features achieved the strongest overall performance for both tasks, suggesting that the handcrafted descriptors still contributed complementary visual information. However, the relatively small improvement over the ResNet-only configuration indicates that the deep embeddings were the primary contributor to classification performance.

4.2 Comparison Between Coarse-Grained and Fine-Grained Classification

The results demonstrated a clear increase in classification difficulty from Task 1 to Task 2. Although Task 1 maintained relatively strong classification performance across most models, Task 2 produced lower validation accuracy and greater difficulty in distinguishing visually similar bird species.

This difference is consistent with the greater visual similarity between bird species, including similar feather textures, colour distributions and body structures. While the coarse-grained animal categories contained more distinctive visual characteristics, the fine-grained bird species differed through more subtle local patterns, making them more difficult to distinguish. Consequently, feature representation became increasingly important for the fine-grained classification task, consistent with the substantial improvement observed when using ResNet18 embeddings.

4.3 Model Performance, Generalisation and Trade-offs

Different models were more suitable for the two classification settings. For the coarse-grained animal classification task, the RBF SVM achieved the strongest overall performance. Its nonlinear decision function provided greater flexibility for modelling complex class boundaries within the high-dimensional combined feature space. Feature standardisation ensured that individual features contributed on comparable scales to the distance-based RBF kernel, while PCA reduced the dimensionality of the feature representation.

Random Forest and CatBoost performed less strongly than the RBF SVM. This difference is consistent with the characteristics of the high-dimensional continuous feature representation, where the axis-aligned partitions used by tree-based models can be less effective than nonlinear decision boundaries for separating complex class structures. In contrast, Logistic Regression remained competitive despite its simpler linear decision boundaries. This suggests that the combined feature representation already provided substantial class separability, while the RBF SVM offered additional flexibility for more complex boundaries.

Although boosting-based and ensemble approaches were competitive, they did not outperform the strongest standalone model. Classifier combination is most effective when the individual classifiers make different errors. In this study, several models produced similar confusion patterns, limiting the additional benefit obtained by combining their predictions.

For the fine-grained bird classification task, Extra Trees was the strongest model among the evaluated classifiers. Its randomisation of candidate features and split thresholds creates diversity across individual trees, allowing the ensemble to capture different patterns within the high-dimensional feature representation. Aggregating predictions across these trees also reduces the dependence on individual tree decisions. Extra Trees additionally required less training and evaluation time than the boosting-based and soft-voting approaches considered in this task, providing a computational advantage.

The consistency of the selected models across cross-validation, holdout and final test evaluation indicates that their performance transferred reasonably across different data splits. Generalisation was less consistent for the fine-grained task, which is consistent with its smaller training set and greater

inter-class visual similarity. These characteristics make performance more sensitive to the composition of the training and validation samples.

A trade-off was also observed between average predictive performance and consistency across validation folds. Some ensemble and boosting-based models showed lower variation across folds, while the selected models achieved stronger average predictive performance. Since mean cross-validation accuracy was the primary model-selection criterion, average predictive performance was prioritised over lower fold-to-fold variation. The additional computational cost of the boosting and soft-voting approaches also did not correspond to consistent improvements over the strongest standalone classifiers.

4.4 Error Analysis

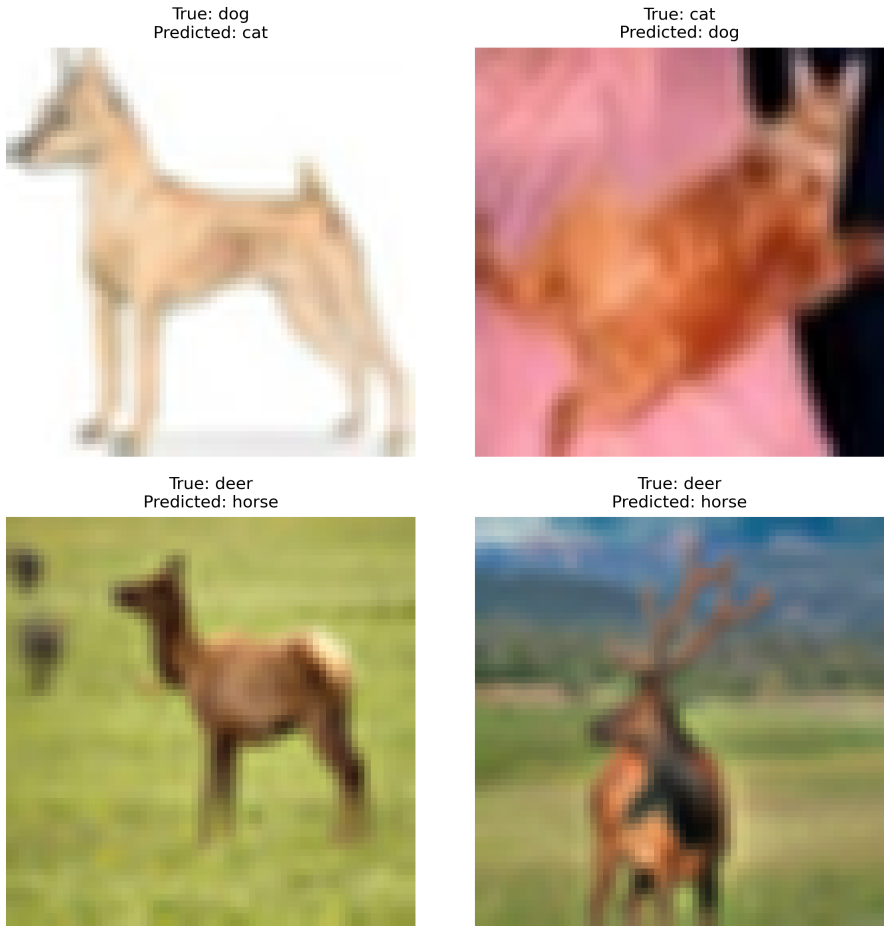


Figure 3: Examples of misclassified Task 1 validation images from the final RBF SVM, including dog–cat and deer–horse confusions.

The misclassification patterns indicate that errors were concentrated among classes with similar visual characteristics. In Task 1, dog and cat formed the most prominent confusion pair, while deer was also repeatedly classified as horse. Examination of individual errors provides further insight into these confusions. As shown in Figure 3, the misclassified examples are dominated by overall animal shape, while the misclassified cat image is noticeably blurred. Since Task 1 images have a resolution of only 64×64 pixels, fine visual details are also limited. These examples suggest that image quality and similarity in overall shape can make the classes more difficult to distinguish.

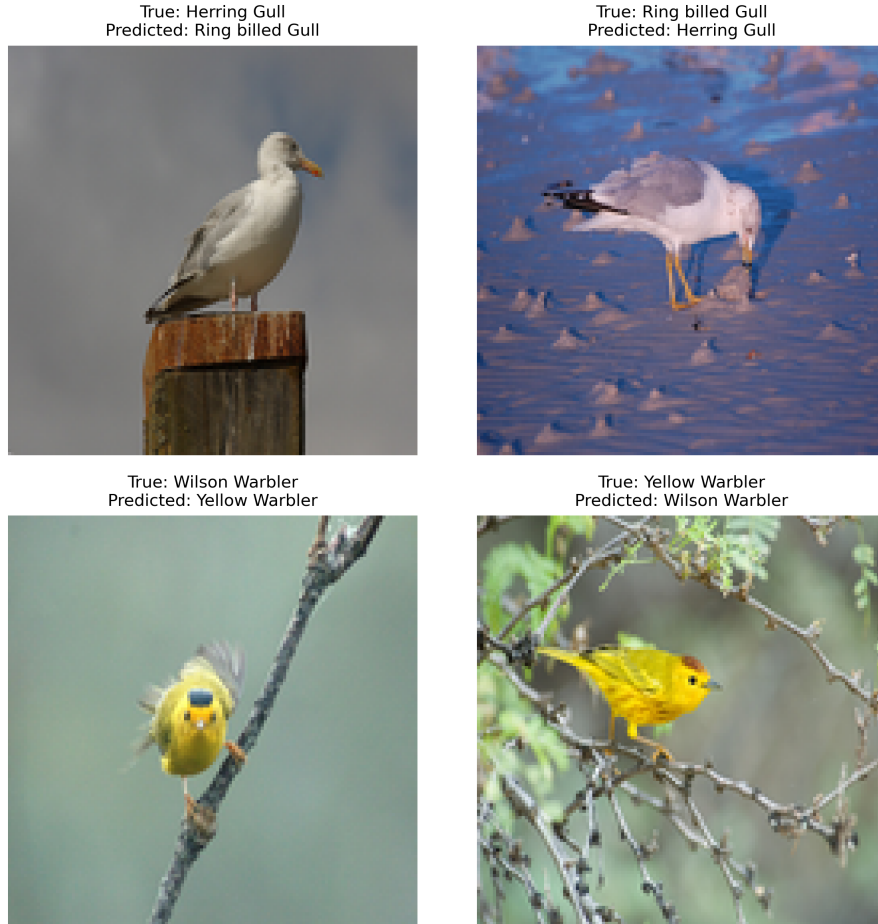


Figure 4: Examples of misclassified Task 2 validation images from the final Extra Trees classifier, showing bidirectional confusion between Herring Gull and Ring-billed Gull and between Wilson Warbler and Yellow Warbler.

The fine-grained errors in Task 2 provide stronger evidence that the classification difficulty depends on subtle local characteristics. As shown in Figure 4, the misclassified Herring Gull and Ring-billed Gull have highly similar overall body structure and predominantly white and grey appearance. Likewise, the Wilson Warbler and Yellow Warbler examples share prominent yellow plumage, while the visual differences between the species occupy relatively small regions of the birds. These examples help explain why broad characteristics such as colour and overall shape are insufficient for reliably distinguishing these species.

Importantly, similar confusion patterns occurred across multiple classifiers, indicating that changing the classifier alone did not resolve the most difficult class pairs. The observed errors are consistent with a limitation of using a fixed global ResNet18 representation for fine-grained classification: species discrimination can depend on local characteristics such as plumage, head markings and bill appearance that occupy only a small part of the image. Image resizing and global feature extraction can reduce the prominence of these details in the final representation. However, the current experiments did not directly examine distances between classes in the ResNet18 embedding space. Therefore, whether the confused species actually form overlapping regions in that feature space remains to be tested.

4.5 Limitations and Future Work

Several limitations should be considered when interpreting the results of this study. The datasets, particularly Task 2, were relatively limited in size, which restricted the amount of variation available for learning visually similar classes. Hyperparameter tuning was also relatively limited and focused only on the strongest-performing models. While this reduced computational cost, more extensive tuning could provide a more comprehensive evaluation of model performance.

The model comparison was also less extensive for Task 2 than for Task 1, as RBF SVM and XGBoost were not evaluated for the fine-grained task. Evaluating a consistent set of classifiers across both tasks would provide a more controlled comparison of model behaviour as classification difficulty increases.

Similarly, the pretrained ResNet18 network was used only as a fixed feature extractor rather than being fine-tuned on the target datasets. Although the extracted embeddings substantially improved classification performance, fine-tuning could allow the representation to better capture task-specific visual characteristics. Further analysis of the ResNet18 embedding space could also determine whether frequently confused classes occupy overlapping regions.

Future work could therefore explore stronger data augmentation, more extensive hyperparameter optimisation and end-to-end deep learning approaches. More advanced architectures, such as Vision Transformers or larger pretrained convolutional neural networks, could also be investigated, particularly for fine-grained image classification.

5 Conclusion

This report investigated two image classification tasks under a coarse-to-fine classification setting using multiple supervised learning approaches and feature representations. The experiments demonstrated that feature representation played a major role in classification performance, with deep visual embeddings extracted from a pretrained ResNet18 model substantially outperforming the provided and engineered features, particularly for the fine-grained bird classification task. Combining the handcrafted and ResNet18 features achieved the strongest feature representation across both tasks.

For the coarse-grained animal classification task, the RBF SVM achieved the strongest overall performance, while the tuned Extra Trees classifier performed strongest among the models evaluated for the fine-grained bird classification task. The error analysis further showed that classification became more difficult when distinctions depended on subtle visual characteristics, particularly between visually similar bird species.

Overall, the experiments highlight the importance of robust feature representations and appropriate model selection for visually complex classification problems. The coarse-to-fine progression also demonstrated the strengths and limitations of different machine learning approaches as classification difficulty increased.

References

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